MIL-B-8621A(Weps)
15 August, 1964
SUPERSEDING
MIL-B-8621(Aer)
20 July 1953

MILITARY SPECIFICATION

BOARDS, CHART, MARK 6A, MARK 6B, AND MARK 6C

This specification has been approved by the Bureau of Naval Weapons, Department of the Navy.

1. SCOPE

1.1 Scope - This specification covers design requirements and all performance requirements for the procurement of the Mark 6A, Mark 6B and Mark 6C chart boards.

1.2 Classification - The chart boards covered by this specification shall be classified as follows:

Mark 6A - Unlighted chart board, 3/4 inch in height.

Mark 6B - Lighted chart board, 3/4 inch in height.

Mark 6C - Lighted chart board, 1-1/4 inches in height, containing large envelope cards.

2. APPLICABLE DOCUMENTS

2.1 General - The following documents, of the issue in effect on the date of the invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

NN-P-515	Plywood, Container Grade
QQ-A-250	Aluminum Alloy Plate and Sheet; General Specification for
QQ-P-416	Plating, Cadmium (Electrodeposited)
PPP-B-566	Boxes, Folding, Paperboard
PPP-B-601	Boxes, Wood, Cleated-Plywood

FSC 6605

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Military

MIL-W-76	Wire and Cable, Hookup, Electrical, Insulated
MIL-A-140	Adhesive, Water-Resistant, Waterproof Barrier-Material
MIL-C-5541	Chemical Films for Aluminum and Aluminum Alloys
MIL-E-5557	Enamel; Heat-Resisting, Glyceryl-Phthalate, Black
MIL-E-5558	Enamel; Wrinkle-Finish, for Aircraft use
MIL-L-6363	Lamps, Incandescent, Aviation Service, General Requirements for
MIL-S-6872	Soldering Process, General Specification for
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series, General Specification for
MIL-P-8685	Primer Coating, Zinc Chromate, Low-Moisture-Sensitivity
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-D-70327	Drawings, Engineering and Associated Lists

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-143	Specifications and Standards, Order of Precedence for the Selection of
MS25237	Lamp, Incandescent, T-1-3/4 Bulb, Midget Flange Base
MS33586	Metals, Definition of Dissimilar

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Naval Aircraft Factory

604555	Computer-Aircraft Navigational, Mark 8-B
605190	Chart Plotting Board Mark 6A

Aeronautical Instruments Laboratory

E1410-0	Mark 6B and Mark 6C Lighted Chart Boards

2.2 Availability of documents - When requesting any of the applicable documents, refer to both title and number. Copies of this specification and applicable documents may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code DCI, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

3. REQUIREMENTS

3.1 Preproduction - The chart boards furnished under this specification shall be products, samples of which have been tested and have passed the preproduction tests specified herein.

3.2 Selection of government documents - Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143, except as provided in 3.2.1 and 3.2.2.

3.2.1 Commercial parts - Commercial parts having suitable properties shall be used, where on the date of invitations for bids, there are no suitable standard parts. In any case commercial parts, such as screws, bolts, nuts, cotter pins, having suitable properties, may be used provided:

(a) They can be replaced by the standard parts (MS or AN) without alteration.

(b) The corresponding standard part numbers are referenced in the parts list and, if practicable, on the contractor's drawings.

3.2.2 Standard parts - With the exception of 3.2.1, MS and AN standard parts shall be used where they suit the purpose. They shall be identified on the drawings by their part numbers.

3.3 Materials - Materials shall conform to applicable specifications and shall be as specified herein. Materials for which there are no applicable specifications, or which are not specifically described herein, shall be of the best quality, of the lightest practicable weight, and suitable for the purpose intended.

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3.3.1 Critical materials - Noncritical materials shall be used where practicable. Where the use of a critical material is essential to meet specification requirements, the material used shall be the least critical of those which are adequate for the purpose.

3.3.2 Nonmagnetic materials - Nonmagnetic materials shall be used for all parts of the chart board except where magnetic materials are essential.

3.3.3 Metals - Metals shall be of the corrosion-resistant type, or shall be suitably protected as specified herein to resist corrosion due to fuels, salt spray or atmospheric conditions to which the chart board may be subjected when in storage or during normal service life.

3.3.3.1 Dissimilar metals - Dissimilar metals as defined in MS33586 shall not be used in intimate contact with each other, unless suitably protected against electrolytic corrosion by means of protective coatings.

3.3.3.2 Aluminum alloy parts - Where practicable, aluminum alloy parts shall be covered with an anodic film conforming to Specification MIL-A-8625. Small holes and case inserts need not be anodized. Aluminum alloys which do not anodize satisfactorily shall be coated with a chemical film in accordance with Specification MIL-C-5541.

3.3.3.3 Iron and steel parts - Iron and steel parts shall be cadmium plated in accordance with Specification QQ-P-416.

3.3.4 Protective treatment - When materials are used in the construction of the chart boards that are subject to atmospheric or environmental conditions likely to cause corrosion in normal service life, they shall be protected against corrosion in a manner that will in no way prevent compliance with the performance requirements of this specification. Finishes and protective coatings which will crack, chip or scale during normal service life or are affected by extremes of atmospheric or environmental conditions, shall not be used.

3.3.5 Fungus-proof materials - Materials which are nutrients for fungi shall not be used where it is practicable to avoid them. Where used, they shall be treated with a fungicidal agent acceptable to the procuring activity.

3.3.6 Fumes and vapors - Materials used in the construction of the chart boards shall not produce corrosive, deleterious, or toxic fumes or vapors under the conditions specified herein.

3.4 Design and construction - The design of the chart board shall be as simple as practicable and shall include all the requirements specified herein. The outline dimensions and general arrangement of the Mark 6A chart board shall conform to NAF Drawing 605190. The outline

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dimensions and general arrangement of the Mark 6B and Mark 6C chart boards shall conform to AIL Drawing E1410-0. The Mark 6A chart board shall consist of a plotting board assembly and a synthetic resin chart board tray. The Mark 6B and Mark 6C chart boards shall consist of a plotting board assembly, a metal chart board tray, a lighting system for illuminating the plotting boards, and accessories as specified herein. The plotting board assembly shall be hinged to the chart board tray, and the board and tray assembly shall be provided with a latch which will form a tight closure. Each Mark 6B and Mark 6C chart board shall be provided with fasteners to secure the chart board to a mount in aircraft. An electric connector for the lighting system shall be provided with each lighted chart board. The chart board shall be so constructed that no parts will work loose in service and will withstand the normal shocks, vibrations and such other conditions as are incident to service without failure, shipping, storage and installation.

3.4.1 Plotting board assembly - The Mark 6A plotting board assembly shall consist of a cover plate and rotatable grid disk. The top surface of the plotting board shall be the cover plate marked with a compass rose scale and a YE-ZB rose. The Mark 6B and Mark 6C plotting board assembly shall consist of a cover plate, rotatable grid disk and backing plate. The top surface of the plotting board shall be the cover plate marked with a compass rose scale, and a YE-ZB rose. The inside surface of the plotting board shall consist of the backing plate, writing pad clips, and a file card and data card holder. The lighted chart boards shall also include a lighting lamp assembly. The grid disk shall be located between the cover plate and backing plate and shall be free to rotate.

3.4.1.1 Cover plate - The cover plate of the Mark 6A chart board shall be made of plastic material suitable for the purpose intended. The cover plate of the Mark 6B and Mark 6C chart board shall be made of plastic material which is sufficiently transparent to render the rotatable grid disk markings clearly visible. The portion of the cover plate which is directly over the light assembly of the Mark 6B and Mark 6C chart boards shall contain a light shield to prevent direct viewing of the lamp filament. This light shield shall be circular and shall be no larger in diameter than the ten mile distance circle on the rotatable grid disk. There shall be two clips on the outside surface of the cover plate to hold a file or data card when in use.

3.4.1.1.1 Compass rose scale - The compass rose scale shall be a circular scale graduated in one degree increments beginning with the numeral 0 (and North Index) placed at the top and on the vertical centerline of the cover plate. The scale graduations shall be accurate to within 1/10 degree.

3.4.1.1.2 YE-ZB rose - The YE-ZB rose shall be marked off in 30-degree segments. The center line of the top segment shall be parallel to the vertical center line of the cover plate.

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3.4.1.2 Rotatable grid disk - The rotatable grid disk for the Mark 6A chart board shall be marked with 1/4 by 1/4 inch squares and other markings as shown on NAF Drawing 605190. The rotatable grid disk for the Mark 6B and Mark 6C chart boards shall be marked with 1/4 by 1/4 inch squares and other markings as shown on AIL Drawing E1410-0. For the lighted chart board, this disk shall be made of suitable plastic material so that it can be satisfactorily edge-lighted together with the cover plate.

3.4.1.3 Backing plate - The backing plate for the Mark 6B and Mark 6C chart boards shall be made of metal and shall be securely attached to the plastic cover plate. Spacers shall be inserted between the cover and backing plate to prevent excessive forces on the rotatable grid disk. The writing pad clips, file card and data card holder, and lighting lamp assembly shall be securely attached to the backing plate.

3.4.1.4 Permanently attached data card - Each chart board shall include a permanently attached data card which shall be durable and of rigid construction, so that it may be pulled out from the front edge of the chart board and present a suitable writing surface. The data card for the Mark 6A chart board shall be marked as shown on NAF Drawing 605190. The data card for the Mark 6B and Mark 6C chart boards shall be marked as shown on AIL Drawing E1410-0. The computer on the data card shall be marked as follows:

> The markings on base -4 of NAF Drawing 604555 shall be duplicated on the data card and a disk conforming to -5 of NAF Drawing 604555 shall be attached to form a Mark 8B computer. For the Mark 6B and Mark 6C chart boards, all nameplate data shall be eliminated from the disk except the marking, "Computer, MK 8B."

3.4.1.5 Detachable data card - The detachable data card for the Mark 6B and Mark 6C chart boards shall be similar to the permanently attached data card except with respect to use and construction. The detachable data card shall be usable when the plotting board is lighted, therefore, it shall be made of material suitable for use as an overlay and shall have light transmitting properties.

3.4.2 Chart board tray - The chart board tray shall support the plotting board assembly. The Mark 6A chart board tray shall include provisions for a pencil. The Mark 6B and Mark 6C chart board tray shall be of aluminum alloy conforming to Specification QQ-A-250, having a tensile strength of 42,000 psi. The chart board tray shall include provisions for pencils, ruler, pencil sharpener, spare lamps, fasteners for attaching the chart board to its mount in the aircraft, and an intensity control for the lighting system of the lighted chart boards. In addition, the Mark 6C chart board tray shall include a holder for five large envelope cards.

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3.4.3 Maintenance - The design of the chart board shall be reasonably simple to facilitate as much as possible disassembly, repair or overhaul, service maintenance, and reassembly using those tools and items of maintenance equipment which are normally available as commercial standards.

3.4.4 Accessories -

3.4.4.1 File cards - Each Mark 6B and Mark 6C chart board shall be provided with a set of file cards consisting of a detachable data card described in 3.4.1.5 and four envelope cards. The file cards shall be housed together and shall be installed to facilitate easy removal and replacement in their stowage compartment. The stowage compartment shall incorporate a locking arrangement to prevent the file cards from inadvertently falling out of the chart board. Each file card shall have a metal-backed identification tab. The tabs shall be so arranged that they are all visible when the chart board is closed and viewed from the top. The center two tabs shall identify the permanently attached data card, and the detachable data card.

3.4.4.1.1 File card envelopes - The file card envelopes for the Mark 6B and Mark 6C chart boards shall consist of a stiffening material and a protective clear plastic cover. The envelope shall accommodate 5 by 8 by .005 inch paper card inserts. Provision shall be made in the stiffening portion of the file card to insure that no portion of the 5 by 8 card insert is obscured by the clips which hold the file card in place on the outer surface of the chart board. The openings of the file card envelopes shall be on the narrow side to insure that the file cards will not slide out when in the normal position. These file card envelopes shall be useable as overlays and shall be capable of transmitting light.

3.4.4.2 Large envelope cards - The Mark 6C chart board only, shall be equipped with five large envelope cards in addition to the file card envelopes described herein. The large envelope cards shall be of construction which is similar to the file card envelopes except that they shall accommodate 8 by 8 by .005 inch paper card inserts. These envelope cards shall be attached to the metal tray of the chart board when in the stowed position.

3.4.4.3 Writing pad - Each Mark 6B and Mark 6C chart board shall be provided with a 5 by 8 inch writing pad containing approximately 40 sheets of paper. The writing pad shall be attached to the backing plate of the plotting board assembly with clips which will hold the pad securely when the pad contains from one to approximately 40 sheets.

3.4.4.4 Pencils - The tray of each Mark 6A chart board shall be provided with one sharpened, eraser-equipped No. 2 lead pencil. The metal tray of each Mark 6B and Mark 6C chart board shall be provided with two sharpened, eraser-equipped No. 2 lead pencils. All pencils shall be securely held in place and easily removable for use.

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3.4.4.5 Ruler - Each Mark 6B and Mark 6C chart board shall be equipped with an attached ruler conforming to AIL Drawing E1410-0. The ruler shall be useable when the chart board is used in daylight or when it is lighted. Both edges of the ruler shall be graduated every 1/8 inch; the scale on one edge shall be marked with numerals as shown on AIL Drawing E1410-0 and the unmarked edge shall have a lusterless finish.

3.4.4.6 Pencil sharpener - Each Mark 6B and Mark 6C chart board shall be equipped with a commercially available pencil sharpener permanently attached to the metal tray in such a manner that access to the sharpener shall be through the front edge of the chart board.

3.4.4.7 Spare lamps - Each Mark 6B and Mark 6C chart board shall be supplied with two spare lamps which shall be held by clips permanently attached to the metal tray.

3.4.4.8 Spare parts - In addition to the file cards installed in each Mark 6B and Mark 6C chart board, there shall be a spare set of five file cards packaged with each Mark 6B and Mark 6C chart board. The Mark 6C chart board shall also have five additional large envelope cards included in each package.

3.4.5 Chart board fasteners - Each Mark 6B and Mark 6C chart board shall be equipped with four fasteners for attaching the chart board to its mount in the aircraft. The location and type of fastener shall be in accordance with AIL Drawing E1410-0.

3.4.6 Weight - The maximum weight of each chart board shall be as follows:

Mark 6A	2.8 pounds.
Mark 6B	3.5 pounds.
Mark 6C	4.0 pounds.

3.4.7 Screw threads - Screw threads 0.060 inch in diameter or larger, shall be in accordance with Specification MIL-S-7742.

3.5 Electrical requirements -

3.5.1 Lighting system - The lighting system for the Mark 6B and Mark 6C chart boards shall be designed to operate on 28 ±1 volts direct current and on the principle of edge lighting to illuminate the plotting area using one lamp centrally located. The light assembly shall be constructed so that the lamp may be easily removed and replaced without the use of tools.

3.5.1.1 Illumination - The lighting system for the lighted chart boards shall provide sufficient illumination of the lighted surface to produce a minimum brightness of 100 micro-lamberts when the lamp is operated at 28 volts. The lamp shall conform to MS25237-327 and shall be red-coated in accordance with the Type II requirements of Specification MIL-L-6363.

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3.5.1.2 Intensity control - The lighting system for the lighted chart boards shall contain a combination on-off switch and wire wound rheostat to control the intensity of the light from zero to maximum brightness.

3.5.1.3 Electric connector - The electric connector for the lighted chart boards shall be in accordance with AIL Drawing E1410-0.

3.5.1.4 Wiring - The wiring used in the Mark 6B and Mark 6C chart boards shall conform to Specification MIL-W-76. All wiring shall be securely and permanently attached to the chart board so as not to interfere with the normal use of the board.

3.6 Markings - All markings shall be durable black and shall be capable of withstanding usage encountered in service.

3.6.1 Cover plate - The compass rose, YE-ZB rose, and name plate shall be marked on the under surface of the cover plate.

3.6.2 Rotatable grid disk - The lettering, numerals, and lines on the rotatable grid disk shall be protected by a layer of transparent vinyl resin .005 inch thick. These markings shall be as readily identified when the board is lighted as when the board is used in daylight.

3.6.3 Data cards - The tab on each of the data cards for the Mark 6B and Mark 6C chart boards only shall be marked "DATA CARD."

3.7 Finishes -

3.7.1 Plotting board assembly - The cover plate shall be durable opaque white outside of the transparent plotting area. This white coating shall be placed on the underside. The upper surface shall be of a durable lusterless finish to take repeated pencil markings and erasures. The upper surface of the backing plate upon which the grid disk rotates shall be anodized and finished in durable high gloss white. The under surface of the backing plate shall be finished in durable lusterless black, as described in 3.7.2.

3.7.2 Chart board - All metal surfaces of the chart board shall be anodized in accordance with Specification MIL-A 8625 and then covered with one coat of zinc chromate primer conforming to Specification MIL-P-8585. All internal metal surfaces shall then be covered with one full wet coat of Type IV instrument black color paint conforming to Specification MIL-E-5557. All external metal surfaces, except as noted in 3.7.1, shall then be covered by one coat of Type I, black wrinkle-finish enamel conforming to Specification MIL-E-5558.

3.7.3 File cards - The upper surface of the protective layer on all file cards as well as the index tabs shall have a lusterless finish to take pencil markings.

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3.8 Performance - The chart board shall perform satisfactorily when subjected to the tests specified in Section 4 of this specification.

3.9 Identification of product - Each Mark 6A chart board shall be permanently and legibly marked in accordance with NAF Drawing 605190. Each Mark 6B and Mark 6C chart board shall be permanently and legibly marked in accordance with AIL Drawing E1410-0. The markings shall not impair the readability of the chart board.

3.9.1 Manufacturer's part number - The manufacturer's part number on the chart board shall be identical with the manufacturer's engineering production drawing number, including applicable dash numbers if the drawing is tabulated and covers more than one part.

3.10 Interchangeability - All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance. Changes in manufacturer's part number shall be governed by the drawing number requirements of Specification MIL-D-70327.

3.11 Instructions - The contractor shall furnish with each chart board one printed copy of instructions in the use of the chart board. These instructions shall contain illustrations and diagrams if necessary and shall be printed on 8-1/2 by 11 inch paper. Prior to printing, two copies of the instructions shall be submitted to the procuring activity for approval.

3.12 Workmanship - The chart board, including all parts and accessories, shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be paid to neatness and thoroughness of soldering, wiring, marking of parts and assemblies, plating, painting, riveting, machine screw assemblage, welding and brazing, and freedom of parts from burrs and sharp edges.

3.12.1 Dimensions - Dimensions and tolerances not specified shall be as close as is consistent with the best shop practices. When dimensions and tolerances affect the interchangeability, operation, or performance of the chart board, they shall be held or limited accordingly.

3.12.2 Fabrication - Machining, drilling, and forming shall be done with the use of accurate templates, jigs, or gages.

3.12.3 Screw assemblies - Assembly screws shall be tight. The word "tight" means that the screws cannot be appreciably tightened without damage or injury to the screw threads or heads.

3.12.4 Riveting - Riveting operations shall be carefully performed in order to insure that the rivets are tight. Rivets heads shall be full size and concentric with the rivet body. Dimensions of the rivets shall be such that the holes will be filled. Loose, cracked, or badly

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formed rivets, including protruding countersunk rivet heads, shall be replaced before acceptance of the chart board. Excess metal shall be removed from countersunk rivet heads as a customary operation. This does not cover eyelets or hollow tubular rivets.

3.12.5 Cleanup - The chart boards shall be thoroughly cleaned of loose, spattered, or excess solder, resin flash which may crumble, and other foreign material after final assembly.

3.12.6 Wire stripping - Wire stripping shall be done without nicking or otherwise damaging the wire.

3.12.7 Soldering - Soldering shall be performed in accordance with Specification MIL-S-6872.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's responsibility - The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification when such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of tests - The inspection and testing of the chart boards shall be classified as follows:

(a) Preproduction tests: Preproduction tests are those tests performed after award of contract on sample chart boards to determine that the chart boards meet the requirements of this specification. The sample chart boards shall be representative in design, performance, and configuration of the chart boards which will be produced on the manufacturer's production time.

(b) Acceptance tests: Acceptance tests are those tests performed on chart boards manufactured and submitted for acceptance under contract.

4.3 Preproduction tests - The preproduction tests of the chart boards shall consist of all of the tests specified under the paragraph headed Test methods.

4.3.1 Preproduction test sample instructions - Preproduction test samples shall consist of three chart boards manufactured in accordance with this specification. The chart boards submitted for preproduction testing shall not have been previously tested except for the

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Individual tests. The samples shall be forwarded to the testing laboratory designated by the procuring activity.

4.3.1.1 Preproduction test sample identification - The test samples shall be plainly identified by securely attached, durable tags marked with the following information:

Sample for Preproduction Testing
BOARDS, CHART, MARK 6A, MARK 6B AND
MARK 6C
Submitted by (Manufacturer's name, date) for
Preproduction Testing in accordance with
Specification MIL-B-8621A(Weps) under contract
or order number _____.
Manufacturer's Part Number

4.4 Acceptance tests - The acceptance tests shall consist of the Individual tests and the Sampling plan tests of this specification. The contractor shall furnish all samples and shall be responsible for accomplishing the required tests. Acceptance tests shall be under the supervision of the Government inspector. The contractor shall furnish test reports showing quantitative results for all tests required by this specification, signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall in no case be construed as a guarantee of the acceptance of the finished product.

4.4.1 Individual tests - Each chart board submitted for acceptance shall be subjected to the Individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, and operational adequacy. As a minimum, each chart board accepted shall have passed the following tests:

Examination of product.
Scale accuracy.

4.4.2 Sampling plan tests - The test samples selected for the sampling tests shall first have passed the Individual tests. When a chart board fails to meet the requirements of this specification, chart boards still on hand or subsequently produced shall not be accepted until the extent and cause of failure have been determined and corrected. In addition, when a failure occurs, shift to one sample out of the first fifteen and proceed as indicated in 4.4.2.1.

4.4.2.1 Sampling plan sample selection - Sampling plan samples shall be selected at random in accordance with the following schedule:

Quantity of Chart Boards Offered for Acceptance		Quantity to be Selected for Testing
First	15	1
Next	50	1

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Quantity of Chart Boards Offered for Acceptance		Quantity to be Selected for Testing
Next	75	1
Next	100	1
Each additional or fraction thereof	200	1

4.4.2.1.1 Sampling plan tests - Each sample selected for Sampling plan testing shall be subjected to the following tests:

Lighting (where applicable).
Computation.

4.5 Test conditions -

4.5.1 Standard conditions - Unless otherwise specified, all tests required by this specification shall be made under the following conditions:

Temperature	Room ambient 25° ±5°C.
Pressure	Normal atmospheric (approximately 29.92 inches Hg.)
Humidity	Room ambient up to 90 percent relative humidity
Input power (for lighted chart boards)	28 ±1.0V DC

4.6 Test methods -

4.6.1 Examination of product - Each chart board shall be examined externally to determine conformance with the applicable drawings and with all the requirements of this specification not covered by tests.

4.6.2 Scale accuracy -

4.6.2.1 Compass rose accuracy - Set one end of the true index successively to 0°, 45°, 90°, and 135°. While on each of these settings read opposite the other end of the true index 180°, 225°, 270°, and 315°, respectively. The error shall be within 1/2 degree.

4.6.2.2 Computer accuracy - Set the 10 graduation of Scale A on the rotating disk opposite the 10 graduation of Scale A on the base of the computer. All graduations on both A scales should then be in alignment. The maximum allowable error at any graduation shall be 0.2 percent of the value of that graduation. With the computer set as indicated above, the index for the density altitude scale should be at the zero foot graduation in the density altitude window. The zero foot pressure altitude graduation should be opposite 15°C at each of the other two computer windows. The maximum allowable limits for each of the last three alignments shall be as follows: Density Altitude Index shall be between ±100 feet. Zero feet pressure altitude at each of the other two computer windows shall appear between 14° and 16° on their respective temperature scales.

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4.6.3 Lighting - The lighting system of the chart board shall be connected to a 28V DC power source and tested to ensure that the lighting meets the requirements of this specification.

4.6.4 Computation - The accuracy of the computer scales shall be tested by the solution of problems of typical atmospheric conditions. A total error of 0.5 percent shall be allowed as specified in Table I.

TABLE I. Computation accuracy

True airspeed computations	Maximum	Minimum
Set 0 ft pressure altitude opposite +15°C; opposite 10 on min scale, read 10 on mile scale	10.05	9.95
Set 10,000 ft pressure altitude opposite 0°C; opposite 315 CAS, read 370 on true A S. scale	371.85	363.15
Set 40,000 ft pressure altitude opposite -60°C; opposite 200 CAS, read 400 on true A.S. scale	402.00	398.00
Density altitude computations		
Set 0 ft pressure altitude opposite +15°C; opposite density altitude scale index, read 0 ft	+5	-5
Set 17,000 ft pressure altitude opposite -10°C; opposite index, read 18,000 ft on density altitude scale	18,100	17,900
Set 40,000 ft pressure altitude opposite -40°C; opposite index, read 41,393 ft on density altitude scale	41,600	41,186
Calibrated altitude computations		
Set 15°C opposite 0 ft pressure altitude; opposite 10 on min scale, read 10 on mile scale	10.05	9.95
Set 40°C opposite 0 ft pressure altitude; opposite calibrated alt 4,600 ft, read 5,000 on correct altitude scale	5,025	4,975
Set -40°C opposite index for pressure altitude; opposite calibrated alt 40,000, read 42,750 on corrected altitude scale	42,964	42,536

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4.7 Rejection and retest - Chart boards which have been rejected may be reworked or have parts replaced to correct the defects and may be resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and action taken to correct the defects shall be furnished the Government inspector.

5. PREPARATION FOR DELIVERY

5.1 Packaging -

5.1.1 Unit packaging - A unit package shall consist of one chart board and spare parts enclosed in a carton conforming to Specification PPP-B-566. The chart boards shall be centered in the carton and supported with die-cut fiberboard pads in such a manner that the edge of the disk or any projections cannot contact the sides of the carton. When the carton is resting on any of its faces, the pads shall support the sides of the chart board in such a manner that the weight of the chart board will not be borne by the edge of the grid disk or any projection.

5.1.2 Intermediate packaging - Unless otherwise specified, 25 unit packages shall be packed in a corrugated or solid fiberboard box.

5.2 Packing - Unless otherwise specified, all items shall receive domestic packing. Shipping containers shall contain the same number of intermediate packages, shall be uniform in size and snugly packed. The gross weight of the fully packed exterior container shall not exceed 200 pounds.

5.2.1 Domestic packing - Intermediate packages shall be overpacked in substantial commercial exterior containers constructed to insure acceptance by common or other carrier for safe transportation at the lowest rate to the point of delivery. Containers shall be able to withstand storage, rehandling and reshipment without the necessity of repacking.

5.2.2 Overseas packing - Unless otherwise specified by the procuring activity, intermediate packages shall be packed in nailed wood or cleated plywood boxes conforming to Specification PPP-B-601. Plywood, if used, shall be in accordance with Specification NN-P-515, Type A or B, Condition I. All boxes shall be furnished with a waterproof case liner with adhesive conforming to Specification MIL-A-140.

5.3 Marking and labeling - Each unit and intermediate package and exterior shipping container shall be marked as specified in MIL-STD-129, including the following nomenclature:

U. S. PROPERTY
CHART BOARD, MARK 6A (or 6B or 6C, as applicable)
SPECIFICATION

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5.3.1 Precautionary marking - The following markings shall appear on two opposite sides of each unit package, intermediate package, and exterior shipping container:

FRAGILE
DELICATE INSTRUMENTS
HANDLE WITH CARE

6. NOTES

6.1 Intended use - The chart boards covered by this specification are intended for use by flight personnel in solving aerial navigation problems and as a convenient place for stowing affiliated equipment.

6.2 Ordering data - Procurement documents should specify the following:

(a) Title, Number and Date of this Specification
(b) The quantity and Mark number of chart boards desired.
(c) Types of packaging and packing desired (See Section 5).

6.3 Drawing requirements - The attention of contractors is invited to the fact that Government contracts for equipment, whether procured directly by the Government or through another contractor, require that all drawings submitted must be in accordance with the requirements of Specification MIL-D-70327.

6.4 Preproduction test provisions - The manufacture of chart boards on contract shall not commence until the samples submitted are pronounced satisfactory by the procuring activity. When a contractor is in continuous production of the chart boards from contract to contract, the submission of further preproduction samples on the subsequent contracts may be waived at the discretion of the procuring activity. Approval of preproduction samples or the waiving of preproduction tests does not preclude the requirement of acceptance testing.

6.4.1 It shall be understood that the chart boards supplied under contract or order shall be identical to the corresponding preproduction sample in design, construction, quality, material, workmanship and method of manufacture. Deviation from the standards of the preproduction sample shall be made only by the procuring activity. Evidence of unauthorized change shall constitute cause for rejection.

6.5 Changes from previous issue - The outside margins of this document have been marked "#" to indicate where changes (deletions, additions, etc.) from the previous issue have been made. This has been done as a convenience only and the government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written irrespective of the marginal notations and relationship to the last previous issue.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

MIL-B-8621A(Wep) BOARDS, CHART, MARK 6A, Mark 6B and M ark 6C

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1426
APR 63

C-8379

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